Problem 1

A random experiment consists of selecting two balls in succession from an urn containing two black balls and one white ball.

- a. Specify the sample space for this experiment
- b. Suppose that the experiment is modified that the ball is immediately put back into the urn after the first selection. What is the sample space now?
- c. What is the relative frequency of the outcome (white, white) in a large number of repetitions of the experiment in part a? In part b?
- d. Does the outcome of the second draw from the urn depend in any way on the outcome of the first draw in either of these experiments?

Problem 2

- a. How many different 7-place license plates are possible if the 2 first places are for letters and the other 5 for numbers?
- b. Suppose now that no letter or number can be repeated in a single license plate, how many different license plates are possible?

Problem 3

Suppose that A and B are mutually exclusive events for which $P(A) = 0.3$ and $P(B) = 0.5$. What is the probability that:

- a. Either A or B occurs;
- b. Both A and B occur.

Problem 4

A total of 8 percent of Canadians smoke ciragettes, 3 percent smoke cigars, and 2 percent smoke both cigars and cigarettes.

- a. What percentage of Canadians smoke neither cigars or cigarettes?
- b. What percentage smoke cigars but not cigarettes?

Problem 5

- a. Prove that $P(EF^c) = P(E) - P(EF)$.
- b. Show that the probability that exactely one of the event E or F occurs is equal to $P(E) + P(F) - 2 \cdot P(EF)$